Total No. of Questions : 8]

SEAT No. :

PA-1506

[Total No. of Pages : 2

[5926]-126
T.E. (Information Technology)
INTERNET OF THINGS
(2019 Pattern) (Semester - I) (314445D) (Elective - I)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) Why 6LoWPAN plays important role in IOT. Explain in detail 6LoWPAN. **[8]**

b) What is advantages of Zigbee? Explain in detail Zigbee protocol stack. **[9]**

OR

Q2) a) Explain Piconet and Scatternet with Bluetooth. **[8]**

b) Explain data aggregation and dissemination in detail. **[9]**

Q3) a) Draw and explain interfacing of output device (Relay) using Ardiuno Uno with program. **[8]**

b) Why the python is the first choice for the Raspberry Pi language than C or C++? **[9]**

OR

Q4) a) Draw and explain interfacing of input device (LED) using Ardiuno Uno with program. **[8]**

b) What is an IOT Device? List different IOT Devices. Explain any 2 devices. **[9]**

Q5) a) Explain Data and message security and Non repudiation and availability with respect to IOT security. **[9]**

b) Explain Python Web Application Framework in detail. Explain How different amazon web services can be used for IOT? **[9]**

OR

P.T.O.

Q6) a) Explain Key elements of IOT Security in details. **[9]**

b) What is threat analysis in IOT? Explain threat analysis in detail. **[9]**

Q7) a) Explain smart city architecture with diagram also state security and privacy challenges in smart transportation in smart city. **[9]**

b) Explain in detail How IOT can be used in home automation? **[9]**

OR

Q8) a) Explain how you will design a smart water management system for agriculture using IOT. **[9]**

b) Explain in detail any two application of health monitoring using IOT. **[9]**

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Total No. of Questions : 8]

SEAT No. :

P-488

[Total No. of Pages : 2

[6003]-709
T.E. (Information Technology)
INTERNET OF THINGS (Elective - I)
(2019 Pattern) (Semester - I) (314445D)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) *Answers Q.1 or Q.2, Q3 or Q4, Q5 or Q6, Q7or Q8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data if necessary.*

Q1) a) What is Bluetooth? Explain the following term of Bluetooth i) Piconet ii) Scatternet. **[9]**

b) Explain in detail the MQTT protocol. **[9]**

OR

Q2) a) Explain Bluetooth protocol architecture. **[9]**

b) What are the development reasons for AMQP? Explain the AMQP model with its main elements. **[9]**

Q3) a) Draw and explain the interfacing of the LED using Raspberry Pi with the program. **[8]**

b) Explain Arduino. What are the different features of Arduino? Discuss digital, analog, and power pins. **[9]**

OR

Q4) a) What is Raspberry Pi? Explain OS installation steps. **[8]**

b) Draw and explain the interfacing of the servo motor using Arduino Uno with the program. **[9]**

P.T.O.

Q5) a) What are different IoT systems vulnerable? Also, explain IoT security challenges. **[8]**

b) What is cloud computing? Explain elements of cloud computing. Also, explain the characteristics of cloud computing. **[9]**

OR

Q6) a) Explain the use and design of a RESTful web API. **[8]**

b) Explain the following key elements of IoT Security: Identity establishment, Access control, Data and message security. **[9]**

Q7) a) Write short note on : **[9]**

i) Home Automation.

ii) Indoor Air quality monitoring.

b) Explain following IoT Applications i) Structural Health Monitoring ii) Smart Parking iii) Smart Road. **[9]**

OR

Q8) a) Write short note on : **[9]**

i) Wearable Electronics.

ii) Smart Payments.

b) Explain how IoT is useful in environment - Weather Monitoring and Noise Pollution Monitoring. **[9]**

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Total No. of Questions : 8]

SEAT No. :

P-7624

[Total No. of Pages : 2

[6180]-144

T.E. (Information Technology)
INTERNET OF THINGS
(2019 Pattern) (Semester - I) (314445D) (Elective - I)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7or Q8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) Explain in detail application Layer Protocols MQTT an4 AMQP. **[8]**

b) Explain Piconet and Scatternet with Bluetooth. Explain advantages and disadvantages of Bluetooth. **[9]**

OR

Q2) a) What is Zigbee? Explain in detail ZigBee protocol stack. **[8]**

b) Explain data aggregation and dissemination in detail. **[9]**

Q3) a) What is an IoT Device? List different IoT Devices. Explain any 2 devices. **[8]**

b) Explain in detail Interfaces (Serial, SPI, I2C). **[9]**

OR

Q4) a) Draw and explain interfacing of input device (LED) using Ardiuno Uno with program. **[8]**

b) Why python is important in IoT? How python plays important role in the Raspberry Pi language than C or C++? **[9]**

Q5) a) Explain Data and message security and Non repudiation and availability with respect to IoT security. **[9]**

b) Explain in detail Web server for IoT and role of Ubidots for Cloud in IoT. **[9]**

P.T.O.

OR

Q6) a) What is threat analysis in IoT? Explain threat analysis in detail. **[9]**

b) Explain Key elements of IoT Security Identity establishment and Access control. **[9]**

Q7) a) Explain in detail IoT Industry Applications - Machine Diagnosis and Prognosis, Indoor Air Quality Monitoring. **[9]**

b) Explain how you will design a smart water management system for agriculture using IoT. **[9]**

OR

Q8) a) What is role of IoT in health monitoring? Explain in detail any two application of health monitoring using IoT. **[9]**

b) Explain in detail How IoT can be used in home automation. **[9]**

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Total No. of Questions : 8]

SEAT No. :

PB-3864

[Total No. of Pages : 2

[6262]-128

T.E. (Information Technology)

INTERNET OF THINGS

(2019 Pattern) (Semester - I) (314445D) (Elective-I)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7or Q8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume Suitable data, if necessary.*

Q1) a) What is Bluetooth? Explain in detail Bluetooth Technology. **[8]**

b) List different application layer protocols? Draw & explain Message Queue Telemetry Transport (MQTT) protocol used in IOT with suitable examples. **[9]**

OR

Q2) a) Enlist & Explain Different IP Based Protocols used in Internet. **[8]**

b) Explain how data Management done in IOT through Data Aggregation and Dissemination methods such as Flooding, Gossiping and SPIN.**[9]**

Q3) a) Differentiate between Arduino and Raspberry? Draw and explain interfacing of LED using Raspberry Pi with the Python code? **[9]**

b) Draw and explain interfacing of Temperature & Humidity sensor (DHT 11) using Arduino with program? **[9]**

OR

Q4) a) What is Raspberry Pi. ? Explain Steps of Installation of OS of Raspberry Pi also Draw and explain interfacing of Switch & LED using Raspberry Pi with the Python code? **[9]**

b) Draw and explain interfacing of DC Motor using Arduino with program? **[9]**

P.T.O.

Q5) a) What is mean by IOT Cloud? Explain Thing Speak Cloud with suitable example. **[8]**

b) Explain different types of possible security issues and attacks of IOT Systems. **[9]**

OR

Q6) a) Differentiate between SaaS, PaaS & IaaS. **[8]**

b) What is mean by IoT Threat modelling? What are the Steps for Threat modeling? Discuss Thread Modeling FRAMEWORKs: STRIDE & DREAD. **[9]**

Q7) a) Describe with suitable diagrams applications of IOT in Smoke/Gas Detector for Home Automation System. **[9]**

b) Explain applications of IOT in Retail Management for Inventory Management, Smart Payments? **[9]**

OR

Q8) a) Explain applications of IOT in smart Agriculture as a Smart Irrigation, Greenhouse Control? **[9]**

b) Describe with suitable diagrams applications of IOT in Smart Fitness & Health Monitoring? **[9]**

Total No. of Questions : 8]

PC-1807

SEAT No. :

[Total No. of Pages : 2

[6353] - 126

T.E. (Information Technology)

INTERNET OF THINGS

(2019 Pattern) (Semester - I) (314445 D) (Elective-I)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Assume Suitable data if necessary.***

Q1) a) Explain in detail IEEE 802.15.4. **[8]**

b) What are disadvantages of IPv4 as compare to IPv6? How IPv6 is more suitable for IoT implementation? **[9]**

OR

Q2) a) Explain how Bluetooth pays important role in communication system? Explain in detail Bluetooth Technology with neat diagram. **[8]**

b) What are advantages of 6LoWPAN? Explain in detail 6LoWPAN technology. **[9]**

Q3) a) What is Arduino? What are the things need to be considered for developing on the Arduino? Which are the official Arduino Boards? **[8]**

b) Draw and explain interfacing of Bluetooth speaker using raspberry pi with python program **[9]**

OR

Q4) a) Compare Arduino Uno & Raspberry Pi Model. **[8]**

b) Draw and explain interfacing of robotic ARM using raspberry pi with python program **[9]**

Q5) a) List different cloud storage models. Explain Cloud Storage models (SaaS, Paas, IaaS) and communication APIs Web server in details. **[9]**

b) Explain how security is important in IoT? Explain the challenges of secure IoT? **[9]**

OR

Q6) a) What is vulnerabilities in IoT? Explain in detail. **[9]**

b) Explain in detail why there is a need of security model in IoT? Explain security model of IoT? **[9]**

Q7) a) Explain in detail how IoT is used in Home automation **[9]**

b) List applications of IoT in agriculture. Explain in detail how IoT can be used in agriculture Applications? **[9]**

OR

Q8) a) Explain in detail how IoT is used in smart city application **[9]**

b) Elaborate on how you will use IoT for remote health care. **[9]**

Total No. of Questions : 8]

SEAT No. :

PD4328

[Total No. of Pages : 2

[6403]-126

T.E. (Information Technology)

INTERNET OF THINGS

(2019 Pattern) (Semester - V) (Elective - I) (314445D)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) Explain in Brief IEEE 802.15.4 protocol. **[8]**

b) Explain Sensor deployment & Node discovery in detail. **[9]**

OR

Q2) a) Explain Application Layer Protocols MQTT and AMQP in short. **[8]**

b) Discuss Wireless Media Access Issues in Internet of Things. **[9]**

Q3) a) Explain Arduino. What are the different features of Arduino? Discuss digital, analog, and power pins of it. **[9]**

b) Draw and explain interfacing of robotic arm using raspberry pi with python program. **[9]**

OR

Q4) a) Draw and explain the interfacing of the DHT11 sensor with Node McU to upload data on cloud with the program. **[9]**

b) Draw and explain interfacing of Bluetooth speaker using raspberry pi with python program. **[9]**

P.T.O.

Q5) a) What is cloud storage? Explain Cloud Storage models (SaaS, PaaS, IaaS) in details. **[8]**

b) Explain the RESTful web API with its designing and development using Django Framework. **[9]**

OR

Q6) a) Write short notes on **[8]**

i) ThingSpeak

ii) Ubidots

b) Explain Challenges for Secure IOT, Threat Modelling in details. **[9]**

Q7) a) Explain how you will design a smart water management system for agriculture using IOT. **[9]**

b) Elaborate on how you will use IOT for remote health care. **[9]**

OR

Q8) a) Explain how you will design an energy management system in a commercial building using IOT. **[9]**

b) Explain how you will design a smart shipment management system using IOT. **[9]**

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Total No. of Questions : 8]

SEAT No. :

PD4325

[Total No. of Pages : 2

[6403]-123

T.E. (Information Technology)
HUMAN COMPUTER INTERACTION
(2019 Pattern) (Semester - V) (314444)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat circuit diagrams should must be drawn wherever necesssary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data if necessary.***

Q1) a) Categorize user profiles for any mobile phone-based money transfer application. (Hint:- Google pay/ PhonePe/ BHIM, etc.) **[7]**

b) Goals are accomplished by methods consisting of operators which are identified by selection rules. illustrate this for following goals. **[10]**

i) To delete a sentence in a graphical text editor

ii) To close window in a graphical text editor

OR

Q2) a) Create a GOMS description of the task of photocopying a paper from a journal. Discuss the issue of closure in terms of your GOMS description? **[7]**

b) Consider designing a user interface for "on-line movie ticket booking" or "hospital admission system". Perform a detailed task analysis. Identify task domain object & actions. Relate task domain & actions with interface objects & actions. **[10]**

Q3) a) A scenario is not only an idealized but also a detailed description of a specific instance of HCI. Scenarios specify how users carry out their tasks in a specified context. Write scenarios for booking an airline ticket from any online application. **[9]**

Note: Generate scenarios to cover a wide range of situations. Include problem situations that will test the system concept.

b) Draw and explain Software design process from HCI perspective. **[9]**

OR

Q4) a) Specify an example for low-fidelity, medium-fidelity, and High-fidelity prototype for any "Exam registration portal". **[9]**

b) Consider any "Online grocery ordering system", draw Model -View-Controller (MVC) framework. Mention the necessary technology solutions available for each of MVC. **[9]**

Q5) a) Explain DECIDE framework with necessary diagram and an example of the same. **[9]**

b) Explain user interface management system (UIMS) with its architecture? **[9]**

OR

Q6) a) What is Usability testing? How will you perform Usability testing on an interactive interface which deals into "Hotel booking". **[9]**

b) Write a short note on- **[9]**

i) Heuristic Evaluation

ii) Cognitive walk through

Q7) a) Explain the following along with real life examples of both. **[10]**

i) Augmented Reality

ii) Virtual Reality

b) Discuss in the detail the Challenges faced by designer while designing interfaces for. **[7]**

i) Smart TV

ii) Smart wrist bands

OR

Q8) a) Draw and explain Design thinking in detail for any suitable application. **[10]**

b) In today's world finding things on web has become very easy. Discuss how multimodal interaction has enriched the experience. Discussion on Input and output through speech, audio, image is expected with relevance to website / products. **[7]**

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